



Getting Started with Research Computing

From Command Line to Job Scheduling

Website: www.rc.colorado.edu

Documentation: curc.readthedocs.io

Support request form: colorado.service-now.com/req_portal?id=ucb_sc_rc_form



Meet the User Support Team



Layla
Freeborn



Brandon
Reyes



Andy
Monaghan



Michael
Schneider



John
Reiland



Dylan
Gottlieb

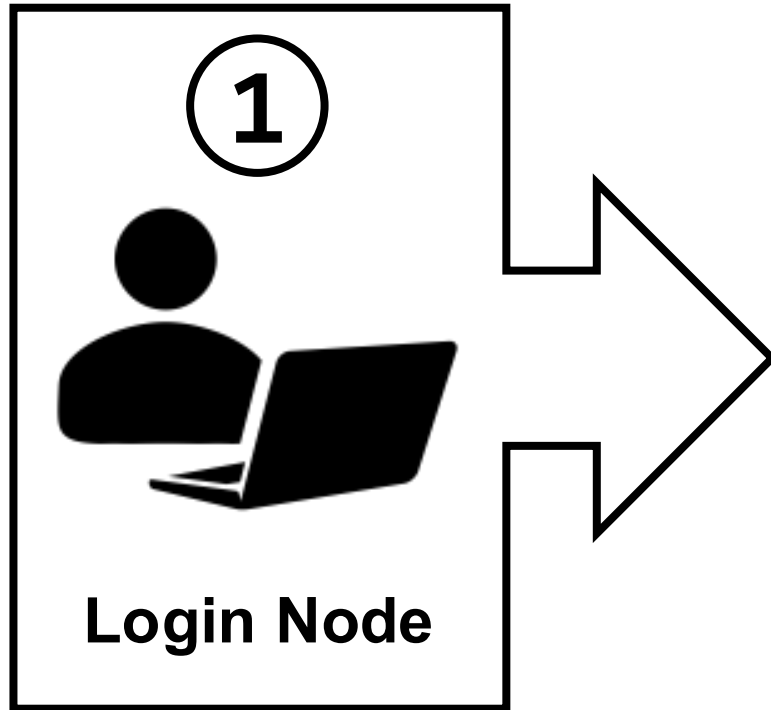


Mohal
Khandelwal

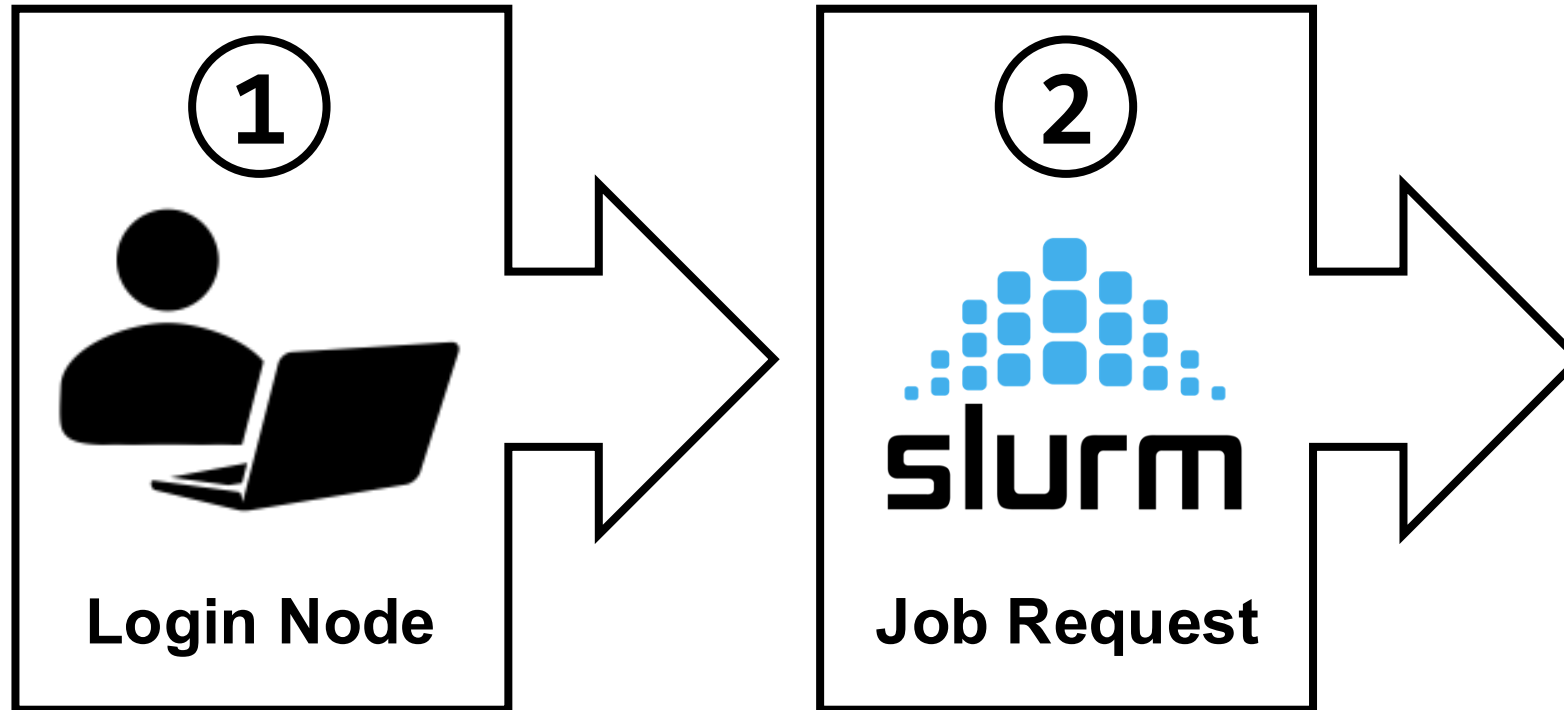


Ask Questions

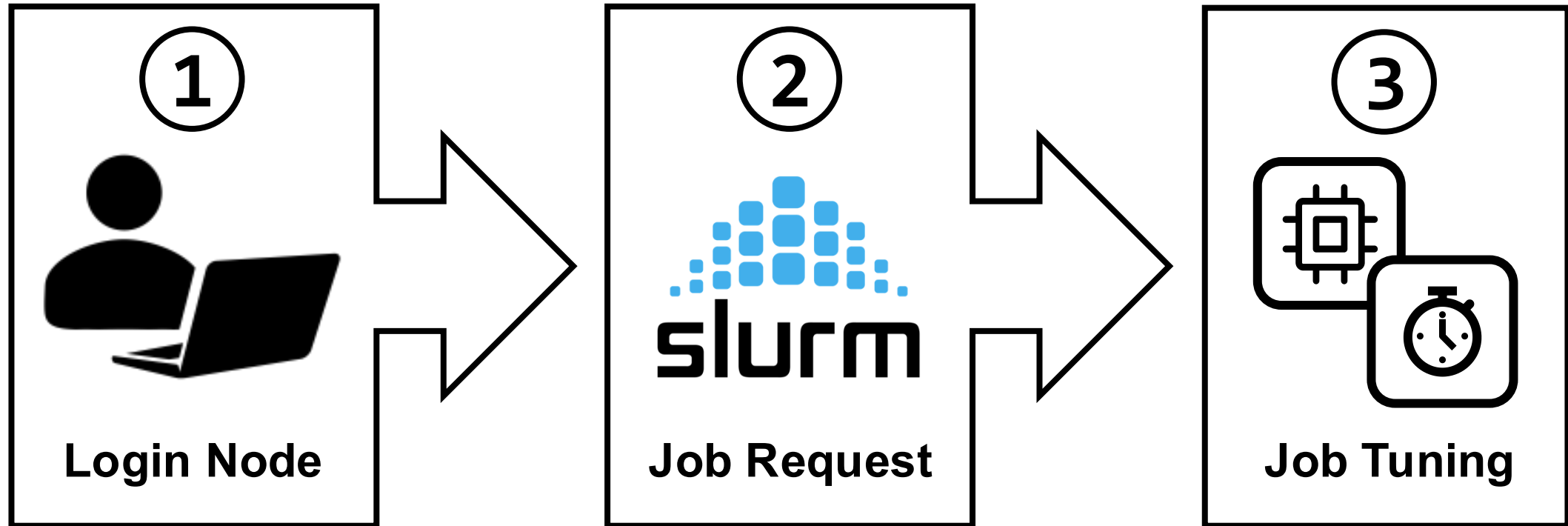
General Overview of Job Submission



General Overview of Job Submission



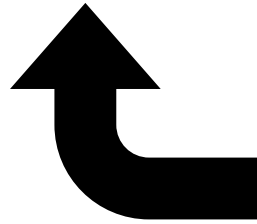
General Overview of Job Submission



Accessing the Terminal

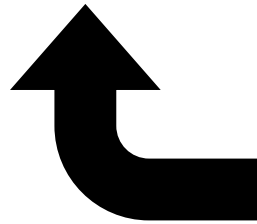


Accessing the Terminal



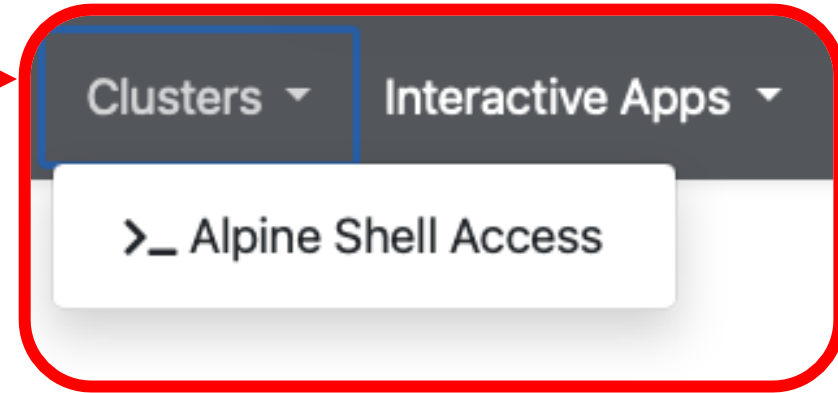
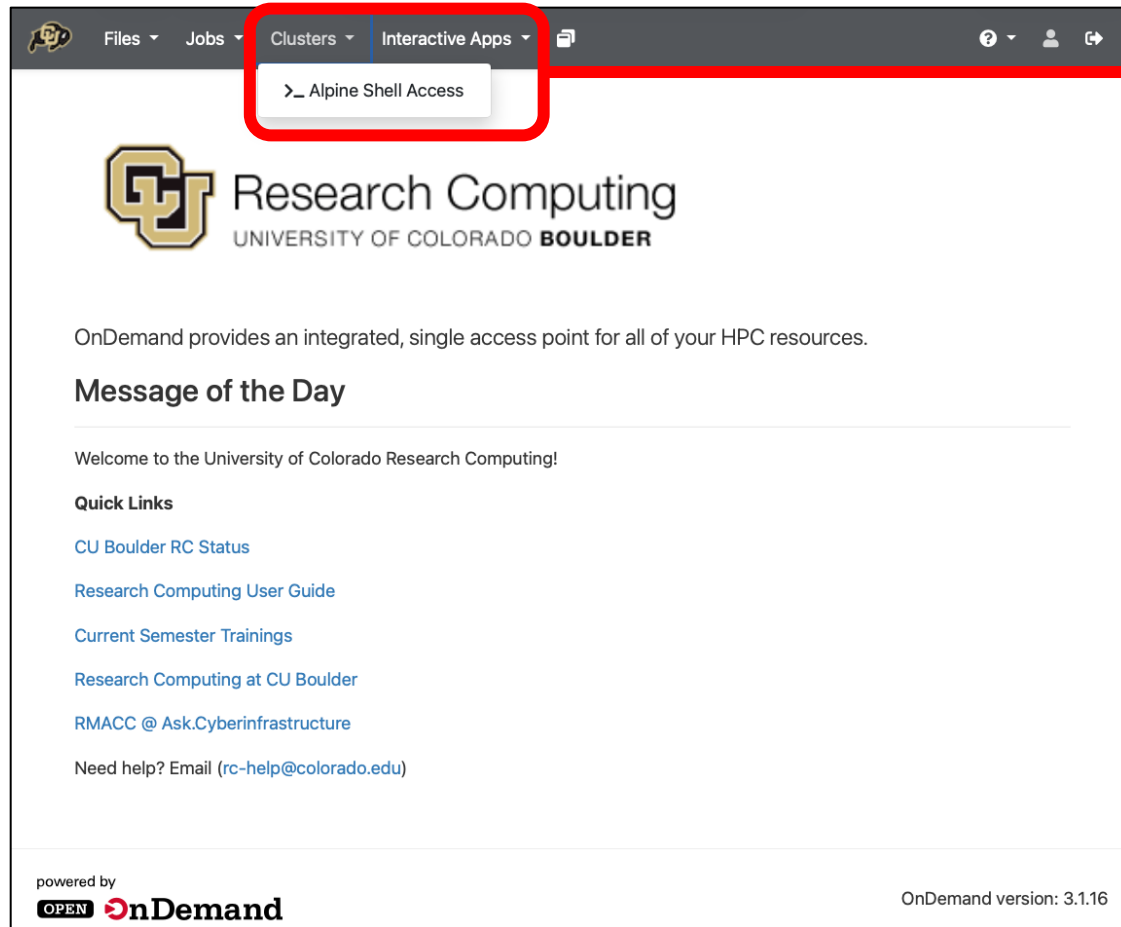
**Available to CU Boulder, CSU,
CU Anschutz**

Accessing the Terminal



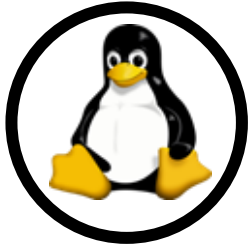
**Available to CU Boulder, CSU,
CU Anschutz**

Accessing the Alpine Shell

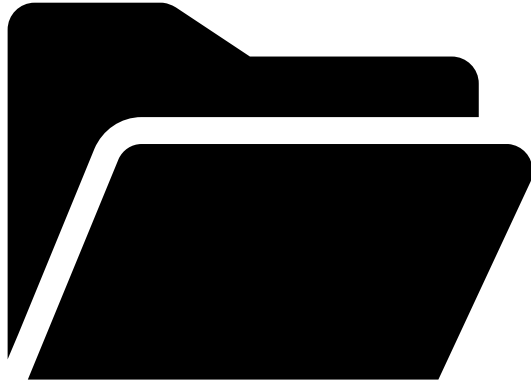


If you would like to access a terminal via ssh,
please see:

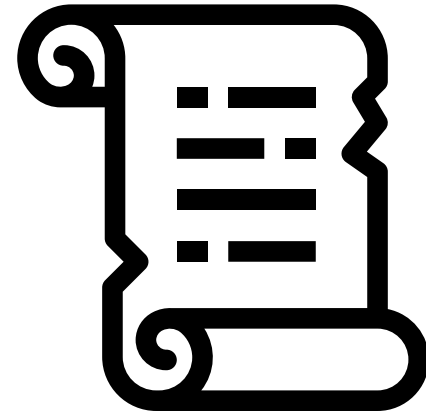
curc.readthedocs.io/en/latest/getting_started/logging-in.html#terminal-access-for-different-operating-systems



Linux Command Line



File Operations



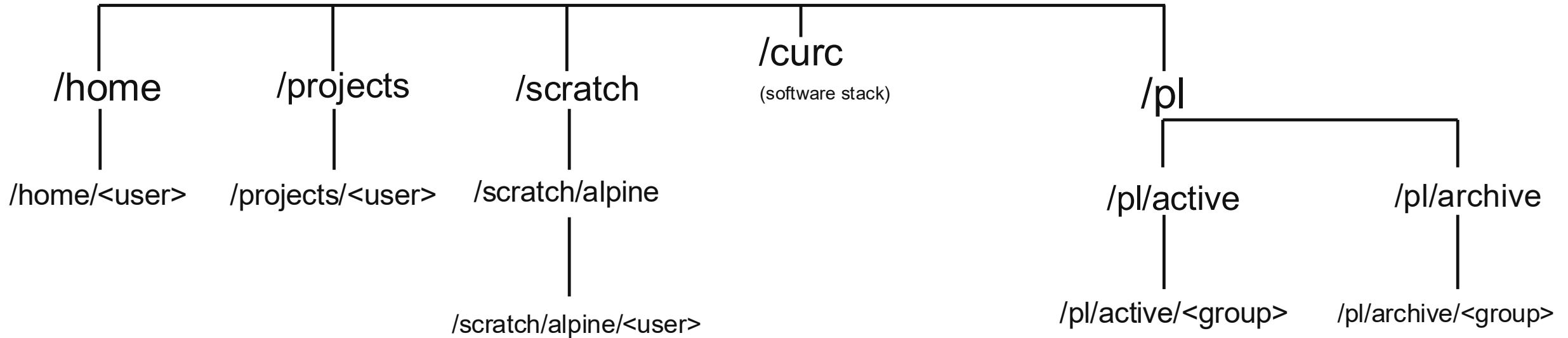
Scripting

List Files and Directories

ls [option] [file/directory]

- l (long format)**
- h (readable file size)**
- S (sort by file size)**
- t (sort by modification time)**
- a (all files)**
- r (reverse sort)**

Standard CURC Directories



File Operations

Change Directory

```
cd [-o] <path>
```

File Operations

Change Directory

```
cd [-o] <path>
```

Make Directory

```
mkdir [-o] <path>
```

File Operations

Change Directory

```
cd [-o] <path>
```

Make Directory

```
mkdir [-o] <path>
```

Copy / Move Files & Directories

```
cp [-o] <source> <dest>
```

File Operations

Change Directory

`cd [-o] <path>`

Make Directory

`mkdir [-o] <path>`

Copy / Move Files & Directories

`cp [-o] <source> <dest>`

Delete Files

`rm [-o] <path>`

Create and edit a file – Terminal Text Editors



Vim

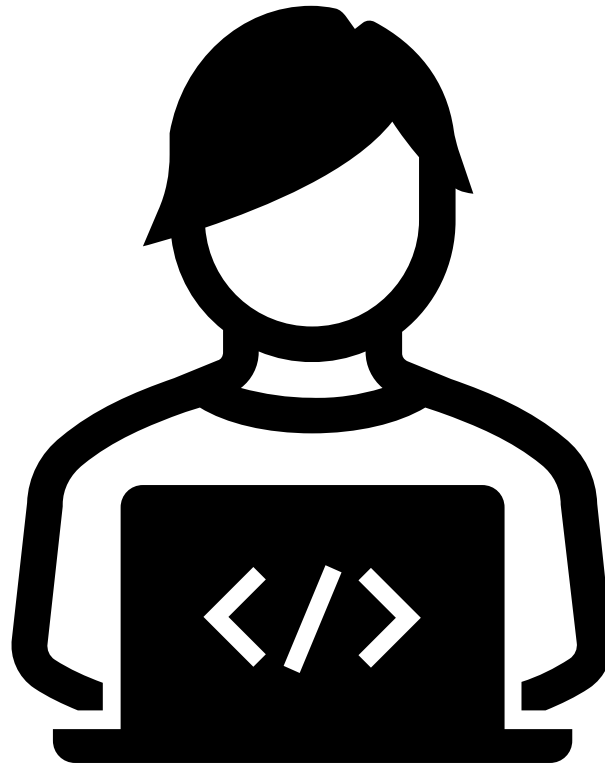
```
      :::  
      iLE88Dj.  :jD88888Dj:  
      .LGitE888D.f8GjjjL8888E;  
      iE  :8888Et.      .G8888.  
      ;i   E888,        ,8888,  
           D888,        :8888:  
           D888,        :8888:  
           D888,        :8888:  
           D888,        :8888:  
           888W,        :8888:  
           W88W,        :8888:  
           W88W:        :8888:  
           DGGD:        :8888:  
                        :8888:  
                        :W888:  
                        :8888:  
                        E888i  
                        tw88D
```

Nano



Emacs

Creating a Bash Script



Shell script basics

- In shell scripts, the first line must specify the shell e.g. `#!/bin/bash`
- The program loader recognizes the `#!` and will interpret the rest of the line (`/bin/bash`) as the interpreter program.
- If a line starts with `#`, it is a comment and is not run.

```
#!/bin/bash
```

```
#look at the files in /tmp
```

```
cd /tmp
```

```
ls
```

Shell to run

Comment

Change directories

List everything in /tmp

Run test.sh

```
#!/bin/bash  
# the files in /tmp.  
cd /tmp  
ls
```

chmod u+x test.sh
./test.sh

Modes (aka permissions)

- View file/directory permissions `ls -l`
- 3 classes of users:
 - User (u) aka “owner”
 - Group (g)
 - Other (o)
- 3 types of permissions:
 - Read (r)
 - Write (w)
 - Execute (x)

`-rwxr-xr--`

Diagram illustrating the permissions string `-rwxr-xr--` with annotations:

- `-`: Directory (indicated by a vertical line below the dash)
- `rwx`: User permissions (indicated by a bracket above the letters and the label "user" above the bracket)
- `r-x`: Group permissions (indicated by a bracket below the letters and the label "group" below the bracket)
- `r--`: Other permissions (indicated by a bracket above the letters and the label "other" above the bracket)

Modes (continued)

`chmod` changes modes:

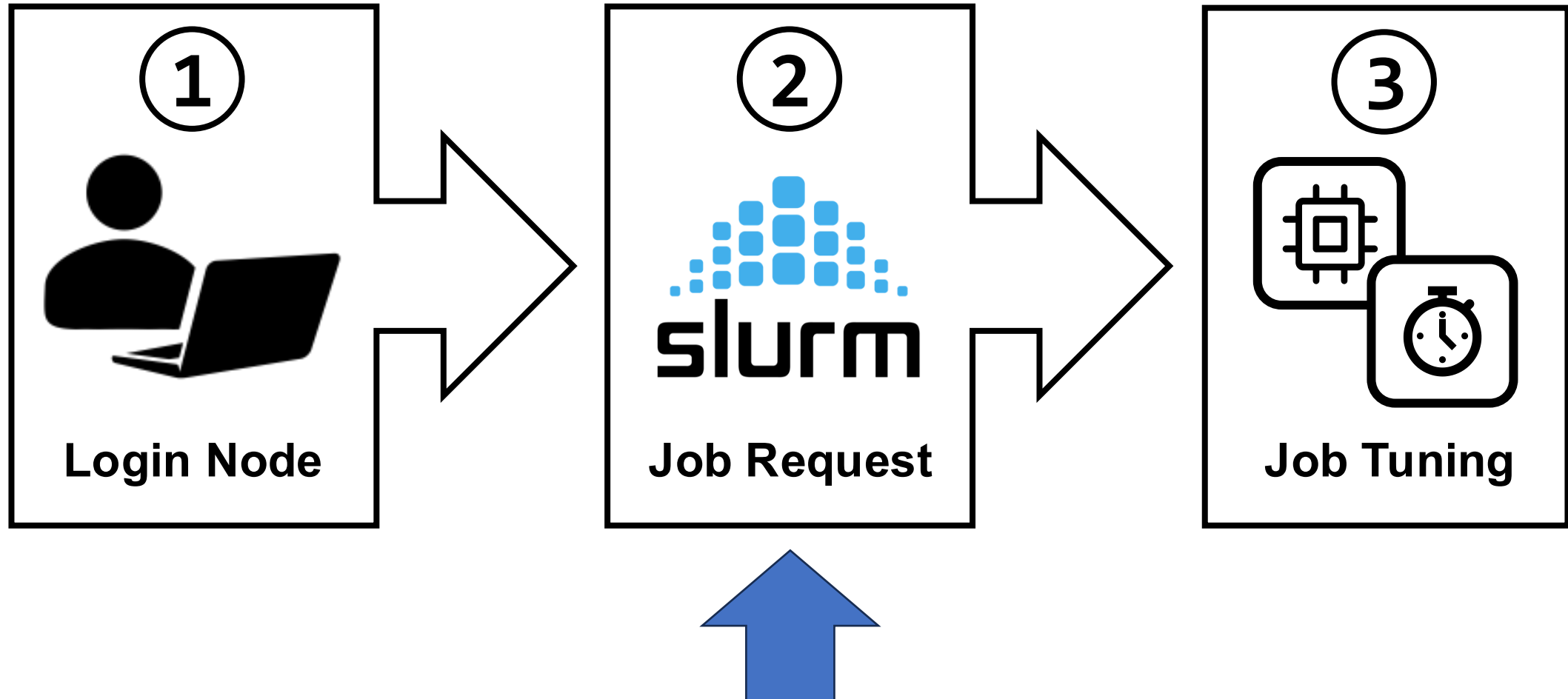
To add write and execute permission for your group:

```
chmod g+wx filename
```

To remove execute permission for others:

```
chmod o-x filename
```

General Overview of Job Submission



Interactive jobs

acompile

- Quick access to resources
- Limit of 4 CPUs for up to 12 hours
- Focused on compiling/building software and small workflows

sinteractive

- Request any partition, more resources, and longer walltime
- Subject to standard wait times (not immediate access)
- As the name implies, allows you to access a job interactively

Running an interactive job

- Here we will run a Python script using an interactive job
- First request resources:

```
$ sinteractive --partition=atesting --qos=testing --time=00:10:00 --ntasks=1
```

- When the job starts you will be put on a compute node and you can execute your commands:

```
$ module load anaconda  
$ python my_very_cool_script.py
```

- To quit:

```
$ exit
```

Batch Jobs

- **Batch Jobs** are jobs you submit to the scheduler that are run later without supervision
- A job script is simply a script that includes **SLURM directives** (resource specifics) ahead of any commands.

Anatomy of a job script

```
#!/bin/bash
```

```
## Directives
```

```
#SBATCH --<option>=<value>
```

```
## Software
```

```
module load <software>
```

```
## User scripting
```

```
<command>
```

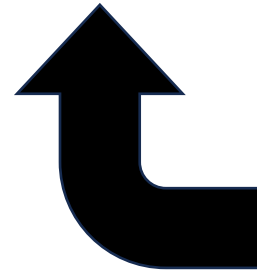


Directives in a job script

```
#SBATCH --<option>=<value>
```

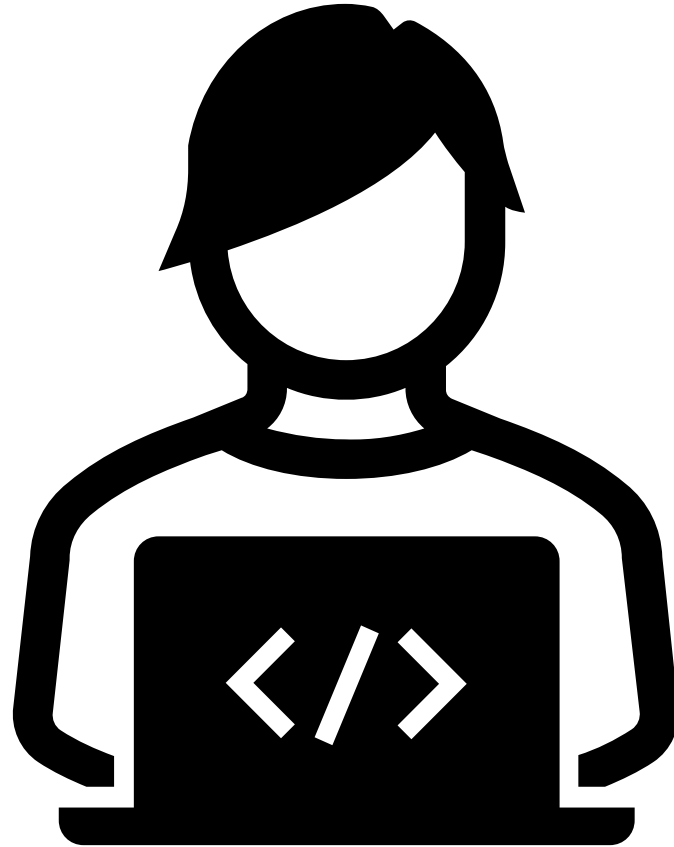
Directives in a job script

#SBATCH --<option>=<value>



**Options listed
at the end**

Demo – Creating a *Batch* Script



Example job script

```
#!/bin/bash

## Directives
#SBATCH --ntasks=1           # Number of requested tasks/cores
#SBATCH --time=0:01:00       # Max run time
#SBATCH --partition=amilan    # Specify Alpine CPU node
#SBATCH --qos=normal          # Specify QoS
#SBATCH --output=test_%j.out  # Rename standard output file

## Software
module purge                  # Purge all existing modules

## User commands
echo "This is a test submitted by $USER"
```



Submitting a Job script

`sbatch <job_file> <other-directives>`

- Command to submit a job to the SLURM job scheduler

Submitting a Job script

`sbatch <job_file> <other-directives>`

- Command to submit a job to the SLURM job scheduler
- If we created the job script “my_first_job.sh” then we would submit it as follows:

`sbatch /path/to/my_first_job.sh`

Submitting a Job script

`sbatch <job_file> <other-directives>`

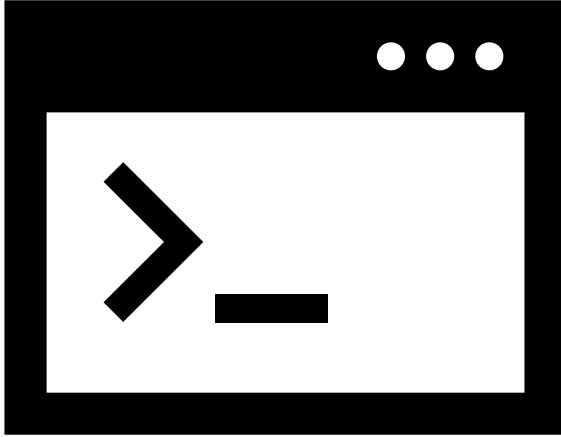
- Command to submit a job to the SLURM job scheduler
- If we created the job script “my_first_job.sh” then we would submit it as follows:

`sbatch /path/to/my_first_job.sh`

- Modify slurm directives in the batch script:

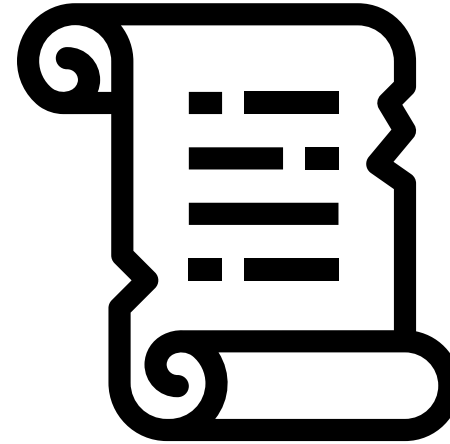
`sbatch /path/to/my_first_job.sh --ntasks=12`

Local vs Environment variables



Environment

**Accessible across
scripts / shells**



Local

**Only exists in
current script / shell**

Environment variables

- Environment variables store important information needed by Linux users and programs
- Type `env` to see your currently set up environment variables
- Useful environment variables:

<code>PATH</code>	directories to search for commands
<code>HOME</code>	home directory
<code>PWD</code>	current working directory
<code>USER</code>	username
<code>LD_LIBRARY_PATH</code>	directories to search for dynamically-loaded libraries

Local vs Environment Variables

```
[username@login-ci3 rc_temp] export g="Hi!"
```

```
#!/bin/bash
```

```
# set local variable
```

```
a="bb"
```

```
# print out the inherited variable
```

```
echo $g
```

Local vs Environment Variables

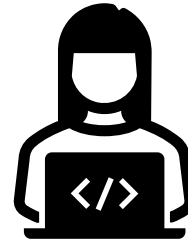
```
[username@login-ci3 rc_temp] ./test.sh
```

```
[username@login-ci3 rc_temp] echo $a
```

Batch vs Interactive



**Batch
Job**



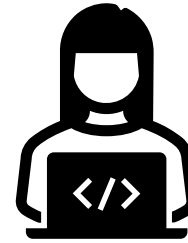
**Interactive
Job**

Batch vs Interactive

- Runs a job script
- No need for user input
- Best for tested workflows and long/big jobs



**Batch
Job**



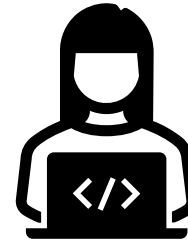
**Interactive
Job**

Batch vs Interactive

- Runs a job script
- No need for user input
- Best for tested workflows and long/big jobs



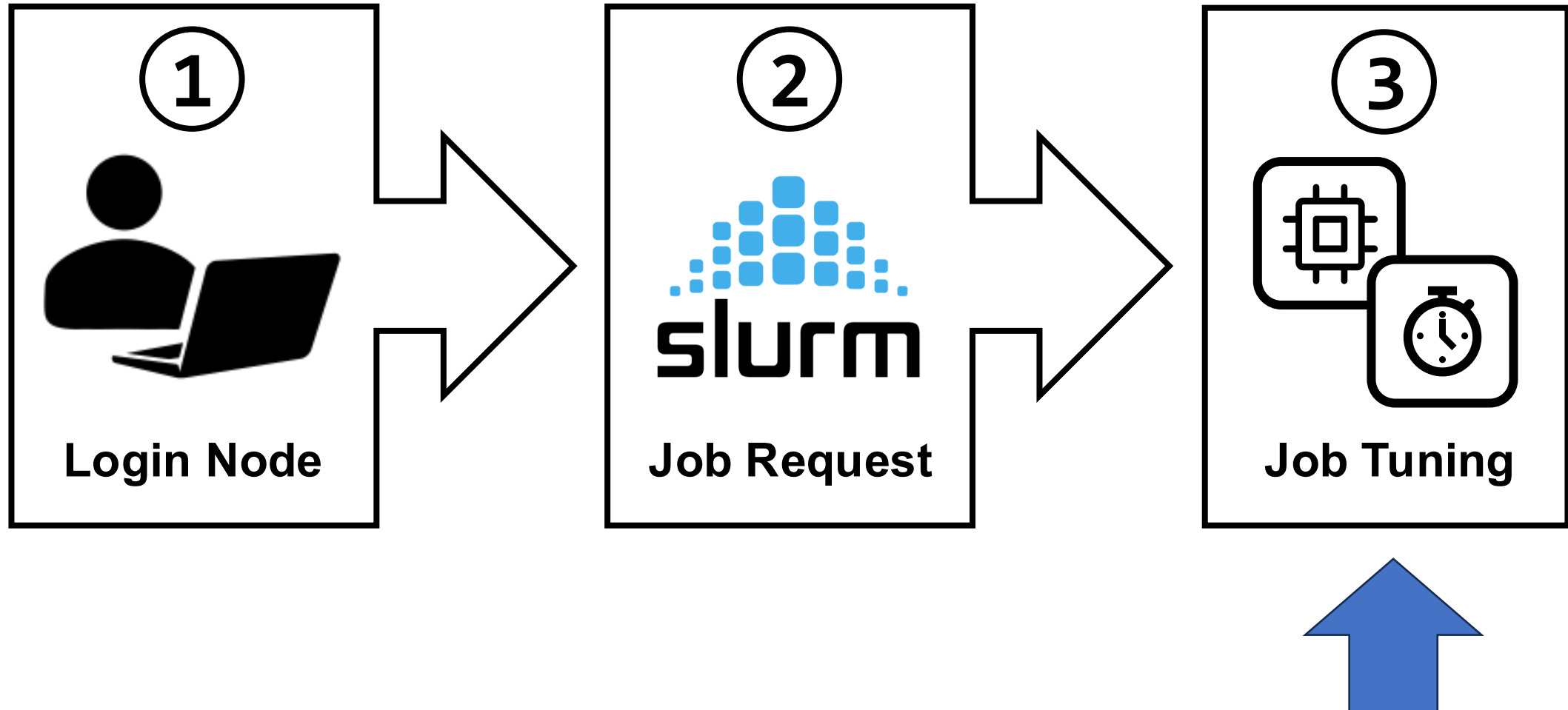
**Batch
Job**



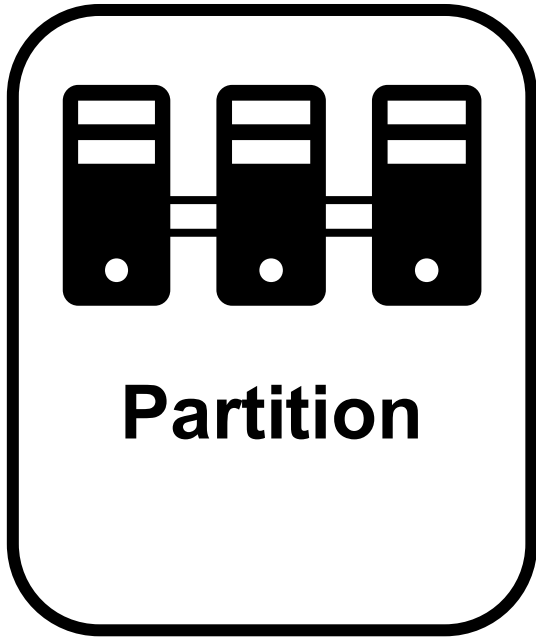
**Interactive
Job**

- Terminal or GUI
- Requires active user input
- Best for development and exploration

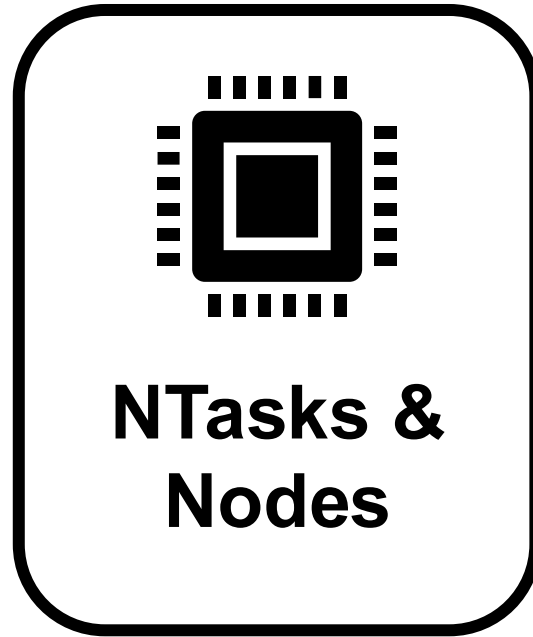
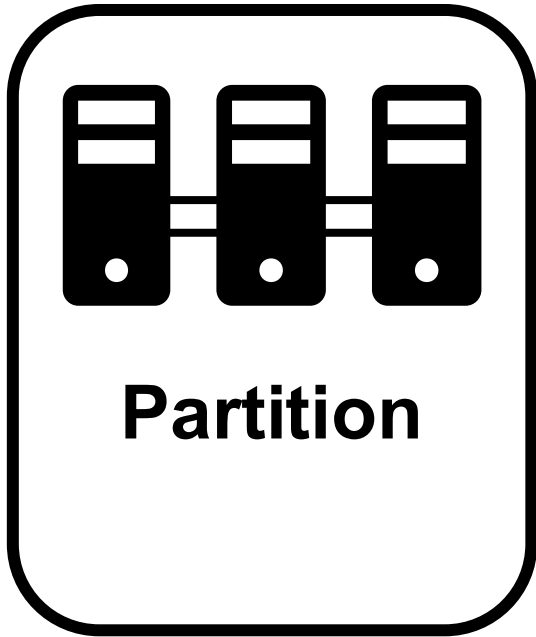
General Overview of Job Submission



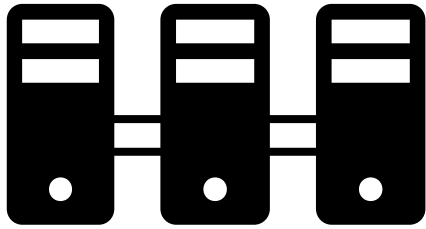
Common Resource Requests



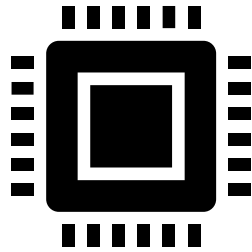
Common Resource Requests



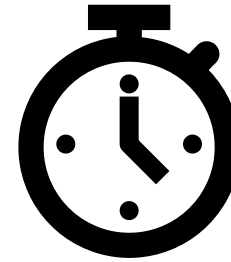
Common Resource Requests



Partition

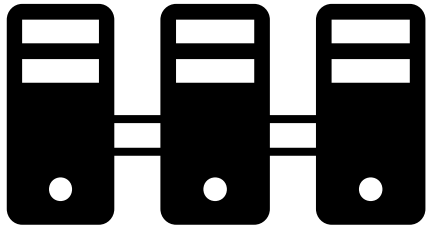


**NTasks &
Nodes**

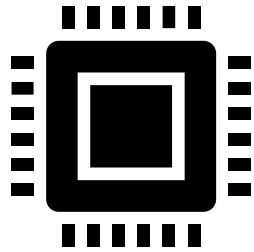


Wall Time

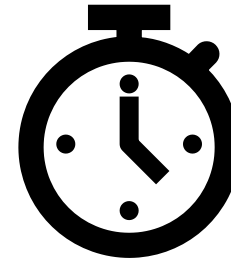
Common Resource Requests



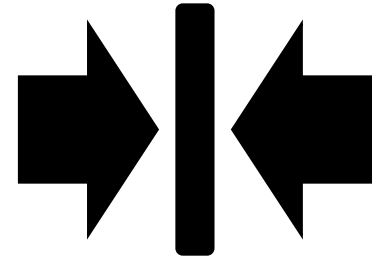
Partition



**NTasks &
Nodes**

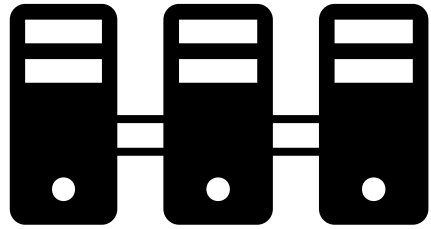


Wall Time

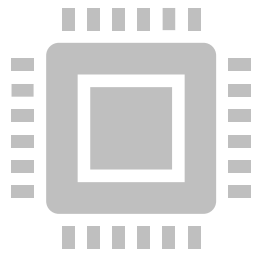


**Quality of
Service**

Common Resource Requests



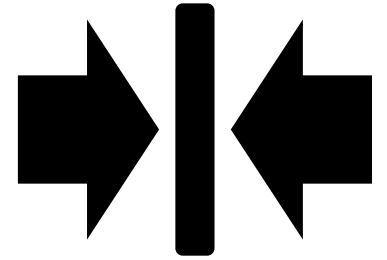
Partition



**NTasks &
Nodes**

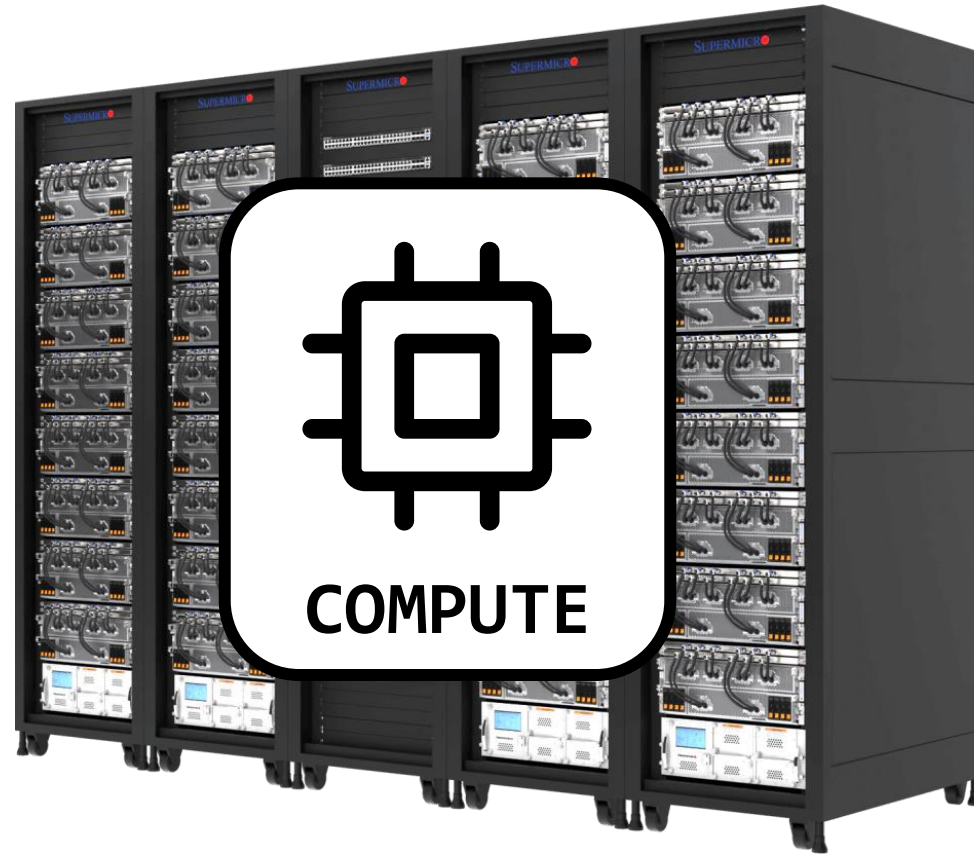


Wall Time



**Quality of
Service**

Alpine Partitions



Alpine Partitions

amilan
General Usage



Alpine Partitions

amilan

General Usage

amem

High Memory



Alpine Partitions

amilan

General Usage

amem

High Memory



aa100

Nvidia GPU's

Alpine Partitions

amilan

General Usage

amem

High Memory



aa100

Nvidia GPU's

ami100

AMD GPU's

Alpine Partitions

- Partitions are a collection of compute nodes e.g. computers with common characteristics

Partition	Description	# of nodes	RAM/core (GB)	cores/node	GPUs/node
amilan	General Compute Node: AMD Milan	406	3.75	32,48,64	0
ami100	GPU Node: 3x AMD MI100	8	3.75	64	3
aa100	GPU Node: 3x Nvidia A100	12	3.75	64	3
amem	High-memory node	24	21.5	48,64	0

Alpine Testing Partitions

- For many of the partitions we have testing partitions that provide quicker access to resources.

Note: there are restrictions on the amount of resources you can request. These restrictions can be found in our documentation.

Partition	Description	# of nodes	RAM/core (GB)	cores/node	GPUs/node
atesting	General Compute Node: AMD Milan	2	3.75	64	0
acompile	General Compute Node: AMD Milan	4	3.75	64	0
atesting_mi100	GPU Node: 3x AMD MI100	1	3.75	64	3
atesting_a100	GPU Node: 3x Nvidia A100	1	3.75	64	6 MIG instances



Quality of Service (QoS)

- Quality of Service specifies additional constraints for a job
 - On Alpine, QoS can be used to run long jobs, specify testing partitions, and select high-memory nodes

QoS	Description	Max wall time	Max jobs/user	Max nodes/user
normal	Default QoS	24 H	1000	128
long	For jobs needing longer wall times	7 D	200	20
mem	High-memory jobs	7 D	n/a	12
testing	For jobs submitted to testing partitions	1 H	1	n/a
compile	For jobs submitted to compile partition	12 H	1	n/a



Common directives

- ★ • **Partition – A collection of compute nodes**
 - `--partition=<partition_name>`
- ★ • **Quality of service (QoS) – System defined constraints for a job (more on this later!)**
 - `--qos=<qos>`
 - **Allocation – Account to “charge to”**
 - `--account=<account_name>`
 - **Number of nodes to run on**
 - `--nodes=<nodes>`
 - **Number of cores to run on**
 - `--ntasks=<number-of-tasks>`
- ★ - Must be specified for every job

Common directives

- ★ • **Wall time** – How long you want to run on these resources
 - `--time=<wall time>`
- **Job name**
 - `--job-name=<jobname>`
- **Output** – Where all output that would be written to the terminal should go
 - `--output=<name>`
- **Send an email when events happen in the job**
 - `--mail-type=<type>`
- **Email address to send updates to**
 - `--mail-user=<user>`

★ - Must be specified for every job

Checking your jobs

- **squeue**: Monitor your jobs status **in queue and while running**:
 - By default, shows all jobs in queue can specify using:

```
$ squeue -u <username>  
$ squeue -p <partition>
```

- **sacct**: Check back on usage statistics of **previous Jobs**
 - By default, only checks all jobs from the start of the current day can specify using:

```
$ sacct -u <username>  
$ sacct --start=MM/DD/YY -u <username>  
$ sacct -j <job-id>
```

Checking your jobs

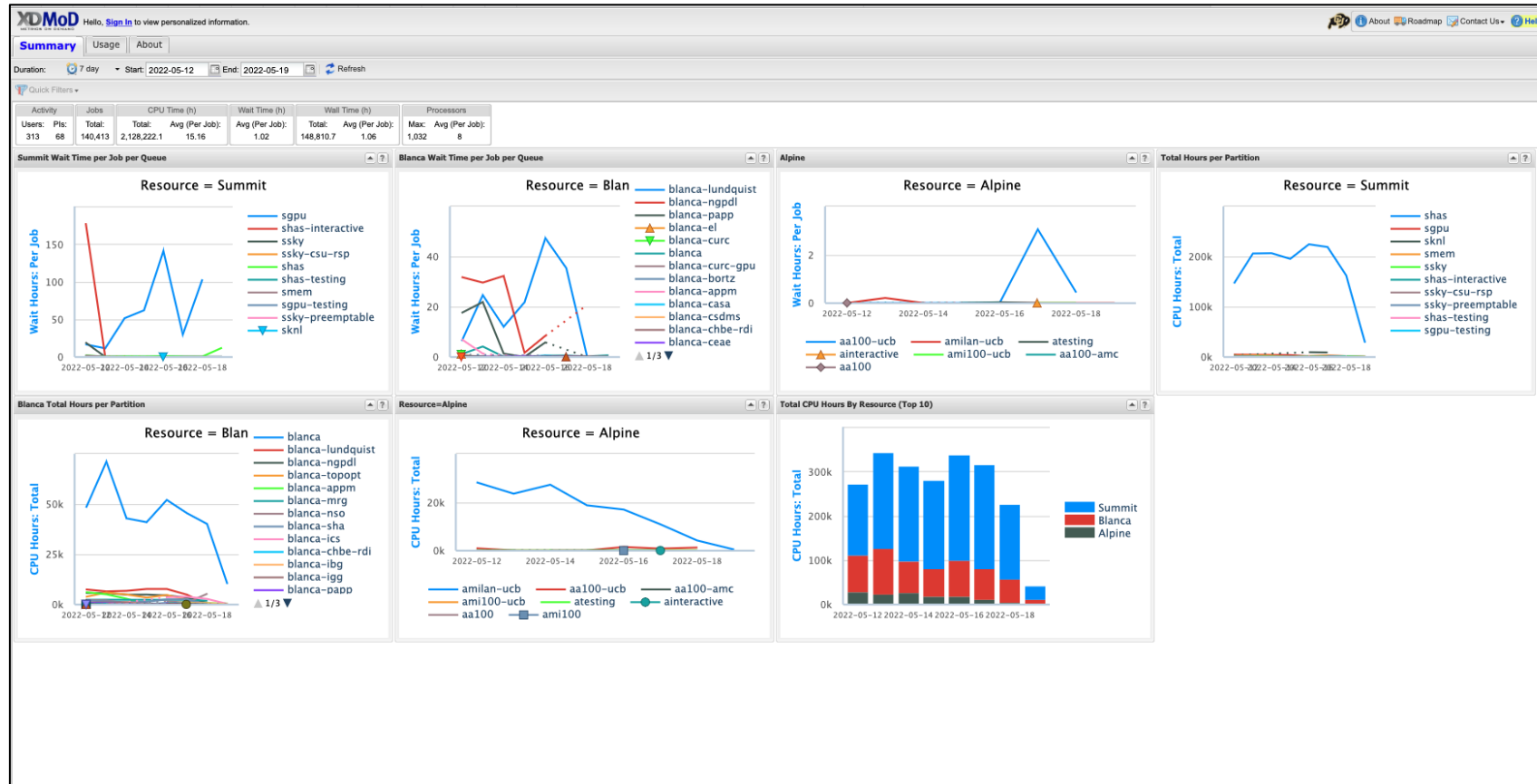
- To check the percentage of CPU and memory usage of a job **after it completes**, use `seff`

```
$ module load slurmttools  
$ seff <job number>
```

- To see general statistics for your submitted jobs, use `jobstats`

```
$ module load slurmttools  
$ jobstats <username> <# of days>
```

Job Monitoring - XDMOD



Thank you!

- **Documentation**: curc.readthedocs.io/
- **Trainings with Center for Research Data and Digital Scholarship (CRDDS)**:
<https://www.colorado.edu/crdds/>
- **Helpdesk**: rc-help@colorado.edu
- **Consult Hours** (Tuesday 12:00-1:00 in-person, Thursday 1:00-2:00 virtually)

Survey and feedback

<https://tinyurl.com/curc-survey18>

